[CMOS] Mini-Project Submission

- Due May 24 at 11:59pm
- Points 0
- Questions 0
- Time Limit None

Instructions

Please take the time to read these instructions carefully. In case of any questions, ask them during the labs (preferably) or send an email to ***jian.ming.huo@vub.be** (mailto:jian.ming.huo@vub.be)* *and* ***jonathan.vrijsen@vub.be** (mailto:jonathan.vrijsen@vub.be)*.

Introduction

During this mini-project, you're asked to design a simple circuit in a 350nm CMOS technology from AMS foundry.

You are provided with the transistor models associated with this technology. Be sure to include this library adequately in every schematic, as described in earlier labs. Failing to include the correct transistor models will lead to an insufficient grade.

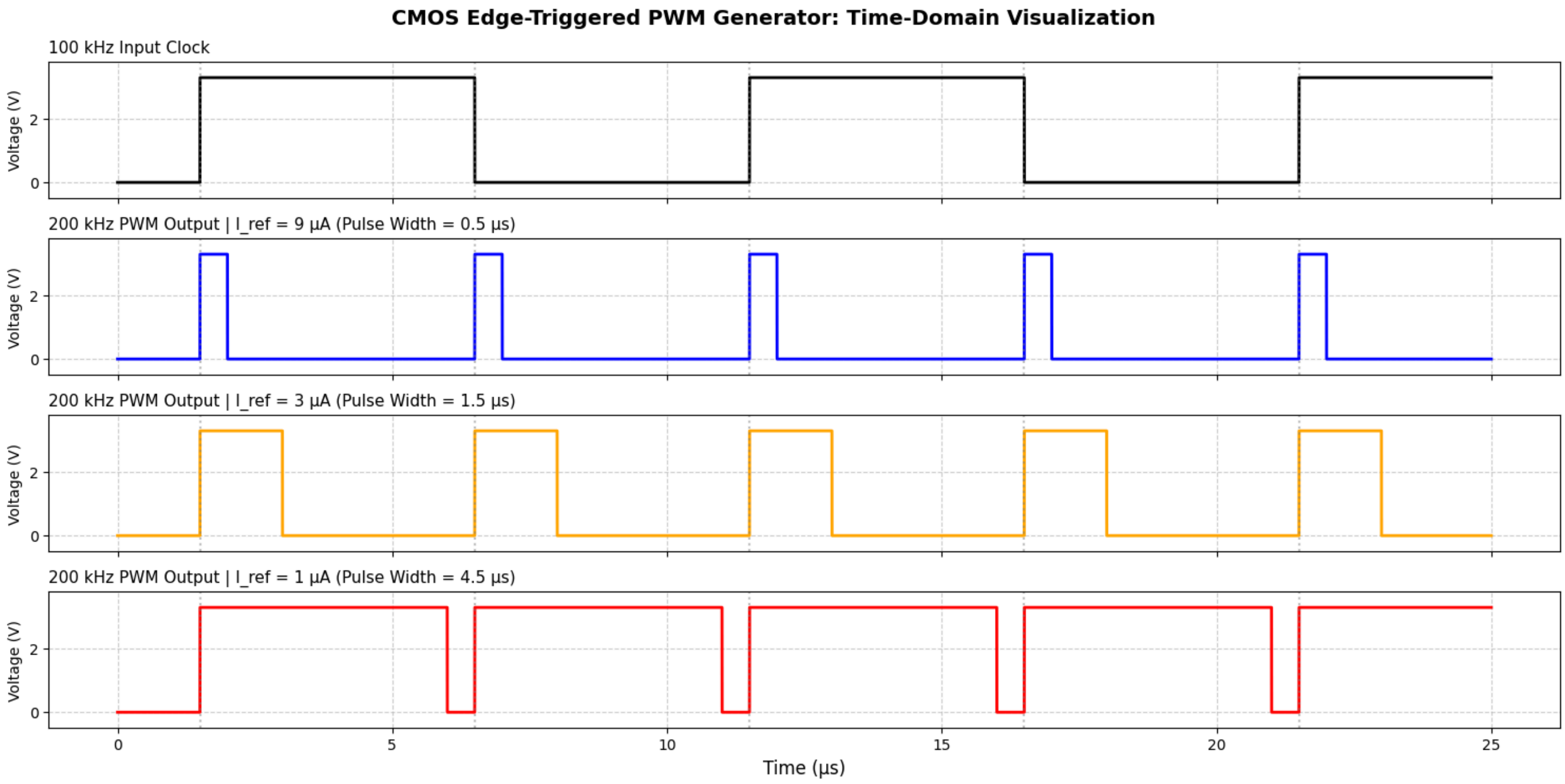
Assignment

The assignment consists out of two parts: a core, compulsory assignment, and additional modules. The core assignment can, if implemented and defended well, only lead to maximum score of 10/20. Students who wish to obtain a higher grade, should add one or more additions.

Some additions can lead to more extra points than others, depending on the design complexity. The maximum extra points an addition can give is indicated as (... / x), where x is the maximally added score. For example, (... / 5) implies that maximal score is increased to 15/20. Additions are cumulative, but the final score is capped at 20/20. (If the grading is still unclear, another example is provided below.)

Core assignment

- You're given a reference current I_{ref} which can vary between 1 μA and 9 μA. Model this as an ideal DC current source with a 10 GΩ output resistance (placed in parallel with the current source). You also have access to a 100 kHz input clock with 10 kΩ output resistance and 5 ns rise/fall time.
- Design an edge-triggered Pulse-Width Modulation (PWM) generator. Your circuit must detect every edge (both rising and falling) of the 100 kHz input clock and generate a high pulse, resulting in a 200 kHz PWM output signal. The output pulse-width is linearly proportional to I_{ref}^{-1}. I_{ref}=9 μA corresponds to a 0.5 μs pulse-width. I_{ref}=1 μA corresponds to a 4.5 μs pulse-width. (See timing diagram below).
- The output signal must drive a 2 pF purely capacitive load, and its rise/fall time (10% -> 90%) should be below 10 ns.
- Verify this linear relationship by performing a parameter sweep: $I_{ref} \epsilon \{1\mu A, 3\mu A, 5\mu A, 7\mu A, 9\mu A\}$.
- Explain which electrical parameters define the output pulse-width. Write this down with the appropriate formules as text in the .asc file.

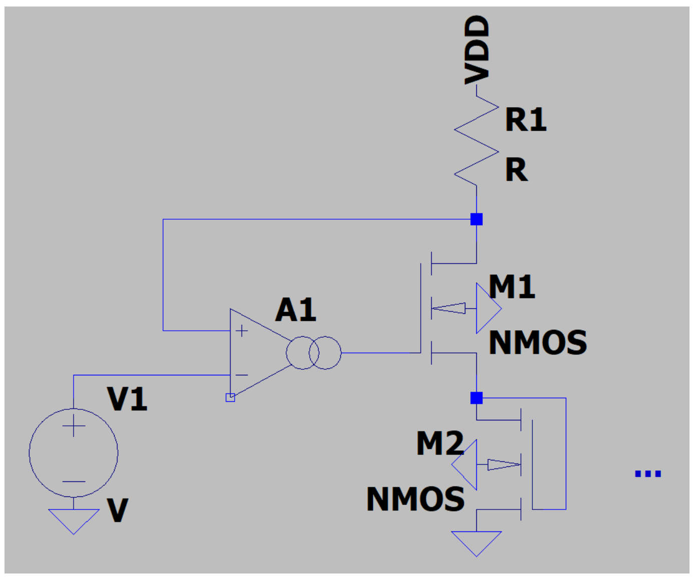


If you implement this circuit well, understand it, and perform well on the defense, you obtain a score of 10/20 on the mini-project.

Be sure to have a look at the provided project lib (Files > Labs > CMOS_project_lib) before starting. It contains, among others, a template to start the core assignment.

Suggested additions

Students who wish to obtain a higher grade should consider implementing one or more of the following suggested additions to the core assignment. Most of these are modular and can as such be combined with one another.

- Linear (large-signal) transconductance amplifier (... / 8)
  - Delete the reference current source from the core assignment. The objective is to make it voltage controlled. Add a DC voltage source (10 kΩ output resistance) which can vary between 1 V and 3.3 V and acts as a control voltage.
  - Generate the reference current based on the control voltage. You should have a highly linear voltage-to-current conversion (so a (source-degenerated) common-source amplifier does not do the job). You may base yourself on (a variation of) this topology:
  - 

  - You obviously need to implement the OTA yourself. To verify its behavior, simulate:
    - Its open-loop low-frequency gain for different input voltages ranging from 0 V to 3.3 V. When and why does this gain degrade?
    - The closed loop phase-margin to prove closed-loop stability.
    - The DC voltage-to-current conversion and illustrate this is indeed linear.
- 90° phase shift based on delay-locked loop (DLL) (... / 10)
  - Create a DLL to generate a 90° phase shifted version of the input clock. By using a DLL, you can obtain a phase shift that is independent of the input clock frequency.
  - The following implementation is suggested:
    - Design a delay line which is a cascade of four voltage-controlled delay elements with a shared control voltage (V_{CP}). The output of the first delay stage will become the 90° phase shifted version of the input clock. The DLL will lock by letting the rising edge of the fourth element's output overlap with the original square wave.
    - As DLL, implement a start-controlled phase-frequency detector and charge pump which generates V_{CP} (if you don't know what it is, Google it!).
      - Use the resettable D-FlipFlop (DFRRLHX1) which is provided to you in the project's template.
    - Also include an asynchronous active low reset (via an additional voltage source which is either high (3.3V) or low (0V) over time).
  - Illustrate the locking over time after reset through a transient simulation.
- Sawtooth generator (... / 4)
  - Use the PWM signal to generate a rising sawtooth signal.
  - The sawtooth signal always spans the voltage range 0V -> 2V, independent of the value of I_{ref}.
- 5-bit Digital-to-pulse-width converter (... / 3)
  - Add five DC voltage sources (output resistance 10 kΩ) which represent a binary digital code. Each of them is either high (3.3 V) or low (0 V).
  - Fix the reference current to 1 μA. Modify the core assignment such that there's a linear relationship between the input digital code and the output pulse width.
  - Verify its behavior for codes "00001", "00010", "00100", and "11111". (small endian notation)
- Voltage-to-pulse-width converter
  - Add a signal DC input voltage (output resistance 10 kΩ) which represents an analog reference voltage $V_{ref} \epsilon [1.5V, 3.1V]$.
  - Fix the reference current to 1 μA. Modify the core assignment such that there's a linear relationship between the output pulse width and $(3.3V - V_{ref})$. $(3.3V - V_{ref}) = 0.2V$ corresponds to a 0.5 μs pulse-width. $(3.3V - V_{ref}) = 1.8V$ corresponds to a 4.5 μs pulse-width.
- Be creative - implement your own ideas (max. ... / 10)
  - If you've got any other ideas or if there's something you would like to try out, discuss this with a TA and add it to your project!

Grading example: If you complete the core assignment (10/20) and add a Sawtooth generator (max +4) and Digital-to-pulse-width converter (max +3), your total possible score becomes 17/20, provided that everything is well implemented and defended.

Design constraints

Your circuit should always comply with the following constraints:

- You can use a ground and a single 3.3V power supply by means of an ideal voltage source.

- All other voltage sources have a 10kΩ series output resistance.
- No other ideal components like voltage or current sources are allowed unless specified.
- Keep the total resistance under 50kΩ and the total capacitance under 5pF (excluding load capacitance), as they require a lot of die area.
- Your circuit should work without initial conditions.
- The output of the circuit should be robust against noise on digital inputs.
- You can only use transistors, resistors, and capacitors.
- Minimum MOSFET length of 0.35μm and minimum width of 0.4μm

Scoring

Scoring of the mini-project will be done based on the following:

- Circuit design
  - Circuit consists out of different functional blocks which each have a clear functionality and need.
  - Every circuit element (MOSFET, passive...) has a clear functionality and need.
  - Design constraints are met.
  - MOSFET sizing is sensible:
    - The use of very wide/long devices (>50 μm) is allowed, but should be justifiable.
  - Circuit should work for the right reasons, in practice:
    - No dependency on simulation settings or numerical considerations.
  - Circuit should be robust to PVT variations and mismatch:
    - It should still operate if parameters change a couple % .
- Project defense
  - Understanding of the larger functional blocks used in the design.
  - Understanding of the function of each MOSFET device. How is it used (current source, amplifier (CS, CG, SF...), switch, resistor...) and biased (triode, saturation)? When does it leave the triode/saturation region?
  - Understanding of the influence of parasitics (parasitic capacitor, channel length modulation...), their influence on the circuits and how to minimize those.
  - Understanding elementary electronic circuits (R-C, C-R, I = C*dV/dt,...).

Deadline and interview

The deadline is **24th of May 2026 @ 23:59 hrs.** Upload a zip folder with your name that contains all files (library.mod, *.asc files, symbols, simulation results …).

Every submitted project has to be **defended orally** with a TA in order to have a grade. There will be an announcement on Saturday 02/05 about the interview.

Take the Quiz